ASSIGNMENT-1

Encrypt / Decrypt using the Pen and Paper Method:

1. Encrypt "CHASE YOUR DREAMS" using

a. Caesar Cipher : For Caesar Cipher, $K = 3$

Formula : $C = (P + 3) \bmod 26$

$C = (2 + 3) \bmod 26 = 5 \bmod 26 = 5 = F$

$H = (7 + 3) \bmod 26 = 10 \bmod 26 = 10 = K$

$A = (0 + 3) \bmod 26 = 3 \bmod 26 = 3 = D$

$S = (18 + 3) \bmod 26 = 21 \bmod 26 = 21 = V$

$E = (4 + 3) \bmod 26 = 7 \bmod 26 = 7 = H$

$Y = (24 + 3) \bmod 26 = 27 \bmod 26 = 1 = B$

$O = (14 + 3) \bmod 26 = 17 \bmod 26 = 17 = R$

$U = (20 + 3) \bmod 26 = 23 \bmod 26 = 23 = X$

$R = (17 + 3) \bmod 26 = 20 \bmod 26 = 20 = U$

$D = (3 + 3) \bmod 26 = 6 \bmod 26 = 6 = G$

$R = (17 + 3) \bmod 26 = 20 \bmod 26 = 20 = U$

$E = (4 + 3) \bmod 26 = 7 \bmod 26 = 7 = H$

$A = (0 + 3) \bmod 26 = 3 \bmod 26 = 3 = D$

$M = (12 + 3) \bmod 26 = 15 \bmod 26 = 15 = P$

$S = (18 + 3) \bmod 26 = 21 \bmod 26 = 21 = V$

$\therefore$ Cipher Text is FKDVH BRXU GUHDPV

b. Shift Cipher with key 7.

Formula : $C = (P + K) \bmod 26$

$C = (P + 7) \bmod 26$

$C = (2 + 7) \bmod 26 = 9 \bmod 26 = 9 = J$

$H = (7 + 7) \bmod 26 = 14 \bmod 26 = 14 = O$

$A = (0 + 7) \bmod 26 = 7 \bmod 26 = 7 = H$

$S = (18 + 7) \bmod 26 = 25 \bmod 26 = 25 = Z$

$E = (4 + 7) \bmod 26 = 11 \bmod 26 = 11 = L$

$Y = (24 + 7) \bmod 26 = 31 \bmod 26 = 5 = F$

$O = (14 + 7) \bmod 26 = 21 \bmod 26 = 21 = V$

$U = (20 + 7) \bmod 26 = 27 \bmod 26 = 1 = B$

$R = (17 + 7) \bmod 26 = 24 \bmod 26 = 24 = Y$

$D = (3 + 7) \bmod 26 = 10 \bmod 26 = 10 = K$

$R = (17 + 7) \bmod 26 = 24 \bmod 26 = 24 = Y$

$E = (4 + 7) \bmod 26 = 11 \bmod 26 = 11 = L$

$A = (0 + 7) \bmod 26 = 7 \bmod 26 = 7 = H$

$M = (12 + 7) \bmod 26 = 19 \bmod 26 = 19 = T$

$S = (18 + 7) \bmod 26 = 25 \bmod 26 = 25 = Z$

$\therefore$ Cipher Text is JOHZL FVBY KYLHTZ

C. ROT 13

Key = 13

Formula : $C = (P + 13) \bmod 26$

$C = (2+13) \bmod 26 = 15 \bmod 26 = 15 = P$

$H = (7+13) \bmod 26 = 20 \bmod 26 = 20 = U$

$A = (0+13) \bmod 26 = 13 \bmod 26 = 13 = N$

$S = (18+13) \bmod 26 = 31 \bmod 26 = 5 = F$

$E = (4+13) \bmod 26 = 17 \bmod 26 = 17 = R$

$Y = (24+13) \bmod 26 = 37 \bmod 26 = 11 = L$

$O = (14+13) \bmod 26 = 27 \bmod 26 = 1 = B$

$U = (20+13) \bmod 26 = 33 \bmod 26 = 7 = H$

$R = (17+13) \bmod 26 = 30 \bmod 26 = 4 = E$

$D = (3+13) \bmod 26 = 16 \bmod 26 = 16 = Q$

$R = (17+13) \bmod 26 = 30 \bmod 26 = 4 = E$

$E = (4+13) \bmod 26 = 17 \bmod 26 = 17 = R$

$A = (0+13) \bmod 26 = 13 \bmod 26 = 13 = N$

$M = (12+13) \bmod 26 = 25 \bmod 26 = 25 = Z$

$S = (18+13) \bmod 26 = 31 \bmod 26 = 5 = F$

∴ Cipher Text is PUNFR LBHE QERNZF

2. Decrypt

a. "WRWDOOB VHFUHW PHVVDJH" using Caesar Cipher.

For Caesar $K = 3$

Formula : $P = (C - K) \bmod 26$

$W = (22 - 3) \bmod 26 = 19 \bmod 26 = 19 = T$

$R = (17 - 3) \bmod 26 = 14 \bmod 26 = 14 = O$

$W = (22 - 3) \bmod 26 = 19 \bmod 26 = 19 = T$

$D = (3 - 3) \bmod 26 = 0 \bmod 26 = 0 = A$

$O = (14 - 3) \bmod 26 = 11 \bmod 26 = 11 = L$

$O = (14 - 3) \bmod 26 = 11 \bmod 26 = 11 = L$

$B = (1 - 3) \bmod 26 = -2 \bmod 26 = 24 = Y$

$V = (21 - 3) \bmod 26 = 18 \bmod 26 = 18 = S$

$H = (7 - 3) \bmod 26 = 4 \bmod 26 = 4 = E$

$F = (5 - 3) \bmod 26 = 2 \bmod 26 = 2 = C$

$U = (20 - 3) \bmod 26 = 17 \bmod 26 = 17 = R$

$H = (7 - 3) \bmod 26 = 4 \bmod 26 = 4 = E$

$W = (22 - 3) \bmod 26 = 19 \bmod 26 = 19 = T$

$P = (15 - 3) \bmod 26 = 12 \bmod 26 = 12 = M$

$H = (7 - 3) \bmod 26 = 4 \bmod 26 = 4 = E$

$V = (21 - 3) \bmod 26 = 18 \bmod 26 = 18 = S$

$V = (21 - 3) \bmod 26 = 18 \bmod 26 = 18 = S$

$D = (3 - 3) \bmod 26 = 0 \bmod 26 = 0 = A$

$J = (9 - 3) \bmod 26 = 6 \bmod 26 = 6 = G$

$H = (7 - 3) \bmod 26 = 4 \bmod 26 = 4 = E$

$\therefore$ Plain text is TOTALLY SECRET MESSAGE

6. "MJQQT BTWQI" using shift Cipher, 5

Key, $K = 5$

Formula: $P = (C - K) \bmod 26$

$M = (12 - 5) \bmod 26 = 7 \bmod 26 = 7 = H$

$J = (9 - 5) \bmod 26 = 4 \bmod 26 = 4 = E$

$Q = (16 - 5) \bmod 26 = 11 \bmod 26 = 11 = L$

$Q = (16 - 5) \bmod 26 = 11 \bmod 26 = 11 = L$

$T = (19 - 5) \bmod 26 = 14 \bmod 26 = 14 = O$

$B = (1 - 5) \bmod 26 = -4 \bmod 26 = 22 = W$

$T = (19 - 5) \bmod 26 = 14 \bmod 26 = 14 = O$

$W = (22 - 5) \bmod 26 = 17 \bmod 26 = 17 = R$

$Q = (16 - 5) \bmod 26 = 11 \bmod 26 = 11 = L$

$I = (8 - 5) \bmod 26 = 3 \bmod 26 = 3 = D$

$\therefore$ Plain Text is HELLO WORLD.

"SJB DTWP YNRJ" using shift cipher, 5

$S = (18 - 5) \bmod 26 = 13 \bmod 26 = 13 = N$

$J = (9 - 5) \bmod 26 = 4 \bmod 26 = 4 = E$

$B = (1 - 5) \bmod 26 = -4 \bmod 26 = 22 = W$

$D = (3 - 5) \bmod 26 = -2 \bmod 26 = 24 = Y$

$T = (19 - 5) \bmod 26 = 14 \bmod 26 = 14 = O$

$W = (22 - 5) \bmod 26 = 17 \bmod 26 = 17 = R$

$P = (15 - 5) \bmod 26 = 10 \bmod 26 = 10 = K$

$Y = (24 - 5) \bmod 26 = 19 \bmod 26 = 19 = T$

$N = (13 - 5) \bmod 26 = 8 \bmod 26 = 8 = I$

$R = (17 - 5) \bmod 26 = 12 \bmod 26 = 12 = M$

$J = (9 - 5) \bmod 26 = 4 \bmod 26 = 4 = E$

$\therefore$ Plain Text is NEW YORK TIME

C. "FRAQ ZBER PBSSRR" using ROT13.

Key, $K = 13$, Formula : $P = (C - 13) \bmod 26$

$F = (5 - 13) \bmod 26 = -8 \bmod 26 = 18 = S$

$R = (17 - 13) \bmod 26 = 4 \bmod 26 = 4 = E$

$A = (0 - 13) \bmod 26 = -13 \bmod 26 = 13 = N$

$Q = (16 - 13) \bmod 26 = 3 \bmod 26 = 3 = D$

$Z = (25 - 13) \bmod 26 = 12 \bmod 26 = 12 = M$

$B = (1 - 13) \bmod 26 = -12 \bmod 26 = 14 = O$

$E = (4 - 13) \bmod 26 = -9 \bmod 26 = 17 = R$

$R = (17 - 13) \bmod 26 = 4 \bmod 26 = 4 = E$

$P = (15 - 13) \bmod 26 = 2 \bmod 26 = 2 = C$

$B = (1 - 13) \bmod 26 = -12 \bmod 26 = 14 = O$

$S = (18 - 13) \bmod 26 = 5 \bmod 26 = 5 = F$

$S = (18 - 13) \bmod 26 = 5 \bmod 26 = 5 = F$

$R = (17 - 13) \bmod 26 = 4 \bmod 26 = 4 = E$

$R = (17 - 13) \bmod 26 = 4 \bmod 26 = 4 = E$

$\therefore$ Plain Text is SEND MORE COFFEE